

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks** (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–14)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

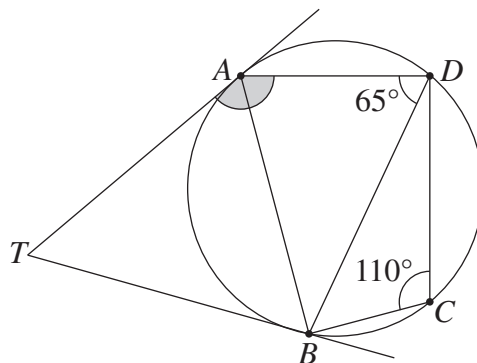
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** Which polynomial is a factor of $x^3 - 5x^2 + 11x - 10$?
- A. $x - 2$
 - B. $x + 2$
 - C. $11x - 10$
 - D. $x^2 - 5x + 11$
- 2** It is given that $\log_a 8 = 1.893$, correct to 3 decimal places.
- What is the value of $\log_a 4$, correct to 2 decimal places?
- A. 0.95
 - B. 1.26
 - C. 1.53
 - D. 2.84

- 3 The points A , B , C and D lie on a circle and the tangents at A and B meet at T , as shown in the diagram.

The angles BDA and BCD are 65° and 110° respectively.

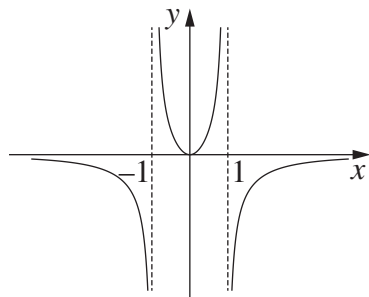


What is the value of $\angle TAD$?

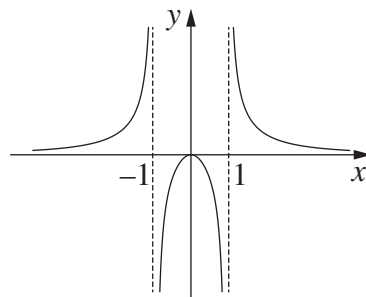
- A. 130°
 - B. 135°
 - C. 155°
 - D. 175°
- 4 What is the value of $\tan \alpha$ when the expression $2 \sin x - \cos x$ is written in the form $\sqrt{5} \sin(x - \alpha)$?
- A. -2
 - B. $-\frac{1}{2}$
 - C. $\frac{1}{2}$
 - D. 2

- 5 Which graph best represents the function $y = \frac{2x^2}{1-x^2}$?

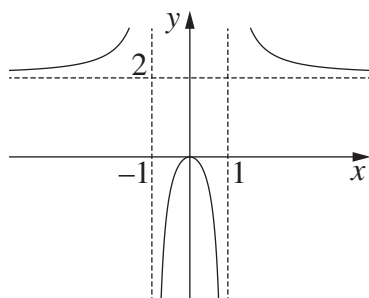
A.



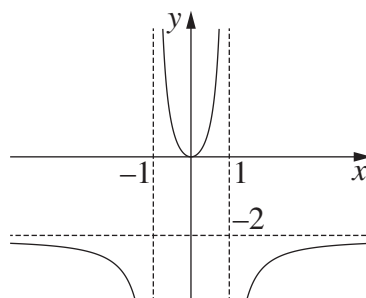
B.



C.



D.

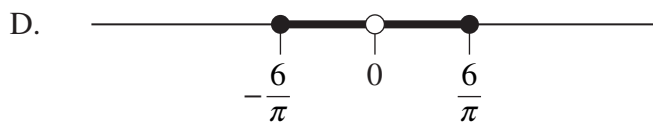
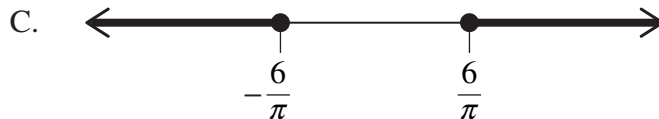


- 6 The point $P\left(\frac{2}{p}, \frac{1}{p^2}\right)$, where $p \neq 0$, lies on the parabola $x^2 = 4y$.

What is the equation of the normal at P ?

- A. $py - x = -p$
 B. $p^2y + px = -1$
 C. $p^2y - p^3x = 1 - 2p^2$
 D. $p^2y + p^3x = 1 + 2p^2$

- 7 Which diagram represents the domain of the function $f(x) = \sin^{-1}\left(\frac{3}{x}\right)$?



- 8 A stone drops into a pond, creating a circular ripple. The radius of the ripple increases from 0 cm, at a constant rate of 5 cm s^{-1} .

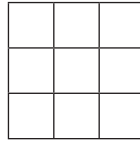
At what rate is the area enclosed within the ripple increasing when the radius is 15 cm?

- A. $25\pi \text{ cm}^2 \text{ s}^{-1}$
 B. $30\pi \text{ cm}^2 \text{ s}^{-1}$
 C. $150\pi \text{ cm}^2 \text{ s}^{-1}$
 D. $225\pi \text{ cm}^2 \text{ s}^{-1}$

- 9 When expanded, which expression has a non-zero constant term?

- A. $\left(x + \frac{1}{x^2}\right)^7$
 B. $\left(x^2 + \frac{1}{x^3}\right)^7$
 C. $\left(x^3 + \frac{1}{x^4}\right)^7$
 D. $\left(x^4 + \frac{1}{x^5}\right)^7$

- 10 Three squares are chosen at random from the 3×3 grid below, and a cross is placed in each chosen square.



What is the probability that all three crosses lie in the same row, column or diagonal?

- A. $\frac{1}{28}$
- B. $\frac{2}{21}$
- C. $\frac{1}{3}$
- D. $\frac{8}{9}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) The point P divides the interval from $A(-4, -4)$ to $B(1, 6)$ internally in the ratio $2:3$. **1**

Find the x -coordinate of P .

- (b) Differentiate $\tan^{-1}(x^3)$. **2**

- (c) Solve $\frac{2x}{x+1} > 1$. **3**

- (d) Sketch the graph of the function $y = 2 \cos^{-1}x$. **2**

- (e) Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} dx$, using the substitution $x = u^2 - 1$. **3**

- (f) Find $\int \sin^2 x \cos x dx$. **1**

Question 11 continues on page 8

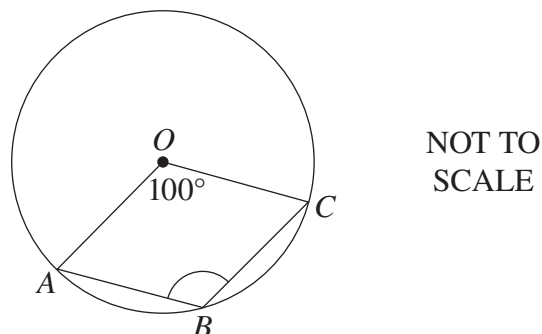
Question 11 (continued)

- (g) The probability that a particular type of seedling produces red flowers is $\frac{1}{5}$. Eight of these seedlings are planted.
- | | | |
|-------|--|----------|
| (i) | Write an expression for the probability that exactly three of the eight seedlings produce red flowers. | 1 |
| (ii) | Write an expression for the probability that none of the eight seedlings produces red flowers. | 1 |
| (iii) | Write an expression for the probability that at least one of the eight seedlings produces red flowers. | 1 |

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The points A , B and C lie on a circle with centre O , as shown in the diagram. **2**
The size of $\angle AOC$ is 100° .



Find the size of $\angle ABC$, giving reasons.

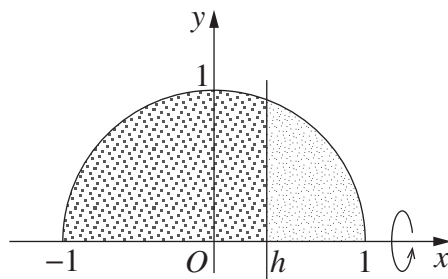
- (b) (i) Carefully sketch the graphs of $y = |x + 1|$ and $y = 3 - |x - 2|$ on the **3**
same axes, showing all intercepts.
- (ii) Using the graphs from part (i), or otherwise, find the range of values of **1**
 x for which

$$|x + 1| + |x - 2| = 3.$$

Question 12 continues on page 10

Question 12 (continued)

- (c) The region enclosed by the semicircle $y = \sqrt{1 - x^2}$ and the x -axis is to be divided into two pieces by the line $x = h$, where $0 \leq h < 1$.



The two pieces are rotated about the x -axis to form solids of revolution. The value of h is chosen so that the volumes of the solids are in the ratio 2:1.

- (i) Show that h satisfies the equation $3h^3 - 9h + 2 = 0$. **3**
- (ii) Given $h_1 = 0$ as the first approximation for h , use one application of Newton's method to find a second approximation for h . **1**
- (d) At time t the displacement, x , of a particle satisfies $t = 4 - e^{-2x}$. **3**

Find the acceleration of the particle as a function of x .

- (e) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$. **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle is moving along the x -axis in simple harmonic motion centred at the origin. **3**

When $x = 2$ the velocity of the particle is 4.

When $x = 5$ the velocity of the particle is 3.

Find the period of the motion.

- (b) Let n be a positive EVEN integer.

(i) Show that $(1+x)^n + (1-x)^n = 2 \left[\binom{n}{0} + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n \right]$. **2**

- (ii) Hence show that **1**

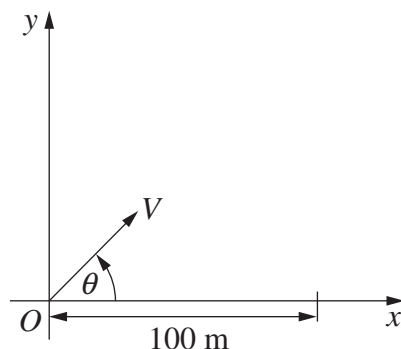
$$n \left[(1+x)^{n-1} - (1-x)^{n-1} \right] = 2 \left[2 \binom{n}{2}x + 4 \binom{n}{4}x^3 + \cdots + n \binom{n}{n}x^{n-1} \right].$$

(iii) Hence show that $\binom{n}{2} + 2 \binom{n}{4} + 3 \binom{n}{6} + \cdots + \frac{n}{2} \binom{n}{n} = n2^{n-3}$. **2**

Question 13 continues on page 12

Question 13 (continued)

- (c) A golfer hits a golf ball with initial speed $V \text{ m s}^{-1}$ at an angle θ to the horizontal. The golf ball is hit from one side of a lake and must have a horizontal range of 100 m or more to avoid landing in the lake.



Neglecting the effects of air resistance, the equations describing the motion of the ball are

$$\begin{aligned}x &= Vt \cos \theta \\y &= Vt \sin \theta - \frac{1}{2}gt^2,\end{aligned}$$

where t is the time in seconds after the ball is hit and g is the acceleration due to gravity in m s^{-2} . Do NOT prove these equations.

- (i) Show that the horizontal range of the golf ball is $\frac{V^2 \sin 2\theta}{g}$ metres. 2
- (ii) Show that if $V^2 < 100g$ then the horizontal range of the ball is less than 100 m. 1

It is now given that $V^2 = 200g$ and that the horizontal range of the ball is 100 m or more.

- (iii) Show that $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$. 2
- (iv) Find the greatest height the ball can achieve. 2

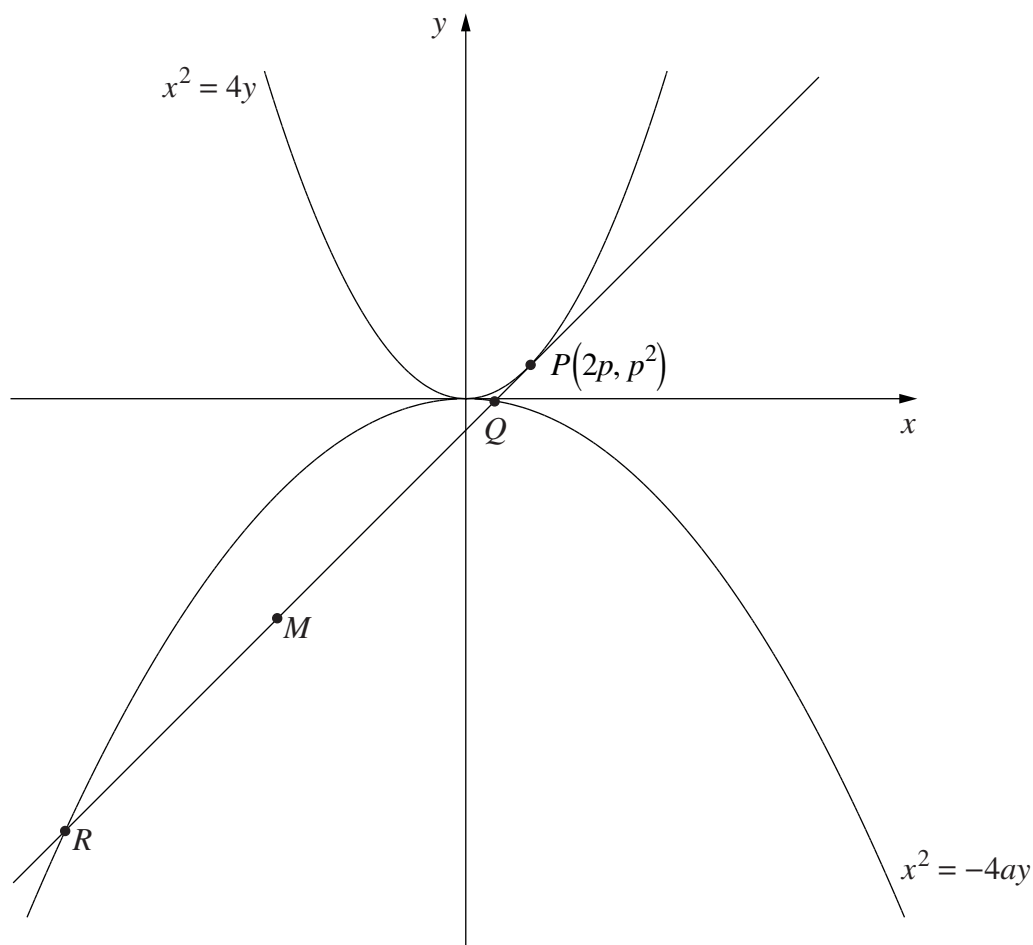
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Prove by mathematical induction that $8^{2n+1} + 6^{2n-1}$ is divisible by 7, for any integer $n \geq 1$. **3**

- (b) Let $P(2p, p^2)$ be a point on the parabola $x^2 = 4y$.

The tangent to the parabola at P meets the parabola $x^2 = -4ay$, $a > 0$, at Q and R . Let M be the midpoint of QR .



- (i) Show that the x coordinates of R and Q satisfy **2**

$$x^2 + 4apx - 4ap^2 = 0.$$

- (ii) Show that the coordinates of M are $(-2ap, -p^2(2a+1))$. **2**

- (iii) Find the value of a so that the point M always lies on the parabola $x^2 = -4y$. **2**

Question 14 continues on page 14

Question 14 (continued)

- (c) The concentration of a drug in a body is $F(t)$, where t is the time in hours after the drug is taken.

Initially the concentration of the drug is zero. The rate of change of concentration of the drug is given by

$$F'(t) = 50e^{-0.5t} - 0.4F(t).$$

- (i) By differentiating the product $F(t)e^{0.4t}$ show that 2

$$\frac{d}{dt}(F(t)e^{0.4t}) = 50e^{-0.1t}.$$

- (ii) Hence, or otherwise, show that $F(t) = 500(e^{-0.4t} - e^{-0.5t})$. 2

- (iii) The concentration of the drug increases to a maximum. 2

For what value of t does this maximum occur?

End of paper

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REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

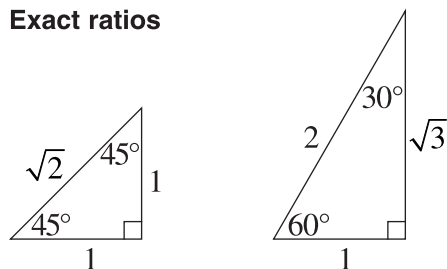
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2017 HSC Mathematics Extension 1

Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	B
3	B
4	C
5	D
6	D
7	A
8	C
9	C
10	B

Section II

Question 11 (a)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 x \text{ value of } P &= \frac{2 \times 1 + 3 \times -4}{2 + 3} \\
 &= -\frac{10}{5} \\
 &= -2
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the chain rule, or equivalent merit	1

Sample answer:

$$\text{Let } y = \tan^{-1}(x^3)$$

$$\begin{aligned}
 y' &= \frac{1}{1 + (x^3)^2} \times 3x^2 \\
 &= \frac{3x^2}{1 + x^6}
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	3
• Correctly identifies 1 and -1 as important, or equivalent merit	2
• Attempts to deal with the denominator, or equivalent merit	1

Sample answer:

$$\frac{2x}{x+1} > 1$$

Multiply both sides by $(x+1)^2$

$$2x(x+1) > (x+1)^2$$

$$2x^2 + 2x > x^2 + 2x + 1$$

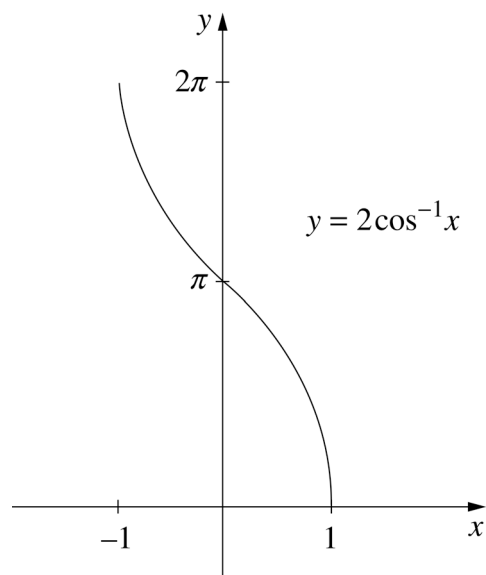
$$x^2 > 1$$

$$\therefore x > 1 \text{ or } x < -1$$

Question 11 (d)

Criteria	Marks
• Provides correct sketch	2
• Indicates correct range, or equivalent merit	1

Sample answer:



Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Provides a correct primitive, or equivalent merit	2
• Attempts to use given substitution, or equivalent merit	1

Sample answer:

$$\int_0^3 \frac{x}{\sqrt{x+1}} dx \quad \begin{aligned} x &= u^2 - 1 \\ \frac{dx}{du} &= 2u \\ dx &= 2u du \end{aligned}$$

$$\begin{aligned} \text{when } x &= 0 & u^2 - 1 &= 0 & u &= 1 \\ \text{when } x &= 3 & u^2 - 1 &= 3 & u &= 2 \end{aligned}$$

$$\int_0^3 \frac{x}{\sqrt{x+1}} dx = \int_1^2 \frac{u^2 - 1}{u} \cdot 2u du$$

$$= 2 \int_1^2 (u^2 - 1) du$$

$$= 2 \left[\frac{u^3}{3} - u \right]_1^2$$

$$= 2 \left[\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right]$$

$$= \frac{8}{3}$$

Question 11 (f)

Criteria	Marks
• Provides correct primitive	1

Sample answer:

$$\int \sin^2 x \cos x dx$$

$$= \frac{\sin^3 x}{3} + c$$

Question 11 (g) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\binom{8}{3} \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^5$$

Question 11 (g) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\left(\frac{4}{5}\right)^8$$

Question 11 (g) (iii)

Criteria	Marks
• Provides correct answer	1

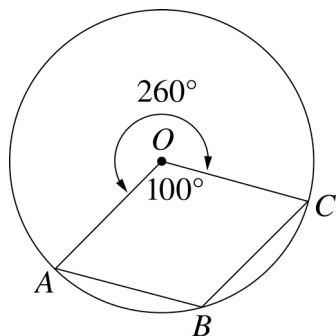
Sample answer:

$$1 - \left(\frac{4}{5}\right)^8$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	2
• Attempts to relate two angles in the diagram, or equivalent merit	1

Sample answer:

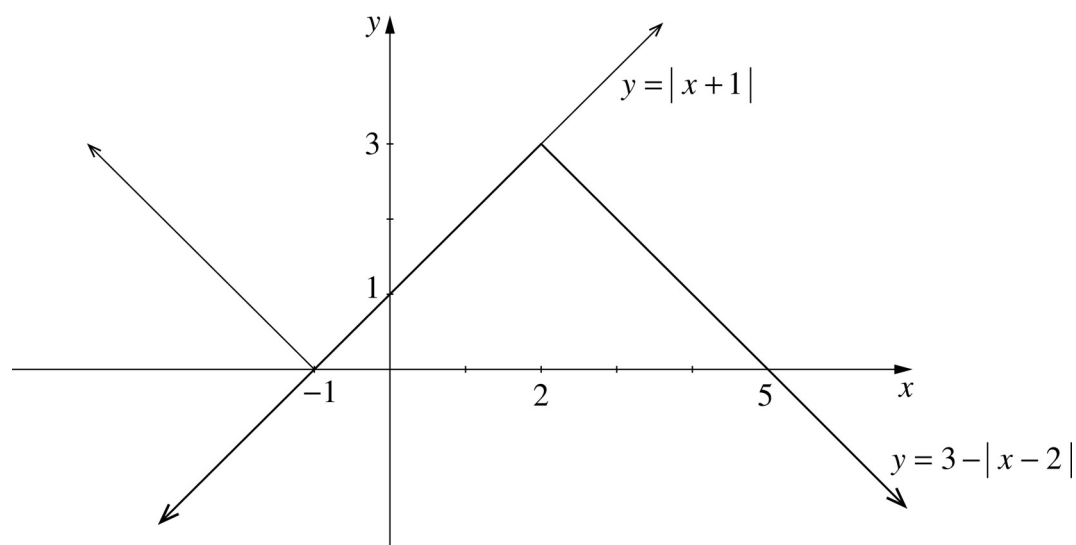


$$\begin{aligned}
 \text{Reflex } \angle AOC &= 360^\circ - 100^\circ && \text{(angle of revolution)} \\
 &= 260^\circ \\
 \angle ABC &= \frac{1}{2} \text{ reflex } \angle AOC && \text{(angle at centre equals twice the angle at} \\
 &&& \text{circumference standing in the same arc)} \\
 &= \frac{1}{2} (260^\circ) \\
 &= 130^\circ
 \end{aligned}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct sketch	3
• Provides correct sketch of $y = 3 - x - 2 $, or equivalent merit	2
• Provides correct sketch of $y = x + 1 $, or equivalent merit	1

Sample answer:



Question 12 (b) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\therefore -1 \leq x \leq 2$$

when $|x + 1| = 3 - |x - 2|$ the graphs have points in common.
ie $|x + 1| + |x - 2| = 3$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Equates the volumes in the correct ratio and attempts to evaluate an integral, or equivalent merit	2
• Provides correct integral for the volume of one solid, or equivalent merit	1

Sample answer:

$$V_1 = \pi \int_h^1 y^2 dx$$

$$= \pi \int_h^1 1 - x^2 dx$$

$$= \pi \left[x - \frac{x^3}{3} \right]_h^1$$

$$= \pi \left[1 - \frac{1}{3} \right] - \pi \left[h - \frac{h^3}{3} \right]$$

$$= \pi \left[\frac{2}{3} - h + \frac{h^3}{3} \right]$$

$$V_2 = \pi \int_{-1}^h 1 - x^2 dx$$

$$= \pi \left[x - \frac{x^3}{3} \right]_{-1}^h$$

$$= \pi \left[h - \frac{h^3}{3} \right] - \pi \left[-1 - \frac{-1}{3} \right]$$

$$= \pi \left[h - \frac{h^3}{3} + \frac{2}{3} \right]$$

Ratio $V_2 : V_1 = 2 : 1 \quad \therefore V_2 = 2V_1$

$$h - \frac{h^3}{3} + \frac{2}{3} = 2 \left(\frac{2}{3} - h + \frac{h^3}{3} \right)$$

$$= \frac{4}{3} - 2h + \frac{2h^3}{3}$$

$$\therefore 3h - h^3 + 2 = 4 - 6h + 2h^3$$

$$\therefore 3h^3 - 9h + 2 = 0$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\text{Let } f(h) = 3h^3 - 9h + 2 \quad h_1 = 0$$

$$h_2 = h_1 - \frac{f(h_1)}{f'(h_1)} = 0 - \frac{f(0)}{f'(0)}$$

$$f(h_1) = f(0) = 2$$

$$f'(h) = 9h^2 - 9$$

$$\begin{aligned} \therefore f'(0) &= 9(0)^2 - 9 \\ &= -9 \end{aligned}$$

$$\begin{aligned} \therefore h_2 &= 0 - \frac{2}{-9} \\ &= \frac{2}{9} \end{aligned}$$

Question 12 (d)

Criteria	Marks
• Provides correct solution	3
• Attempts to use $\ddot{x} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$, or equivalent merit	2
• Correctly finds $\frac{dt}{dx}$, or equivalent merit	1

Sample answer:

$$t = 4 - e^{-2x}$$

$$\frac{dt}{dx} = +2e^{-2x}$$

$$\therefore \frac{dx}{dt} = \frac{1}{2}e^{2x}$$

$$v = \frac{1}{2}e^{2x}$$

$$v^2 = \frac{1}{4}e^{4x}$$

$$\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$= \frac{d}{dx}\left(\frac{1}{8}e^{4x}\right)$$

$$= \frac{4}{8}e^{4x}$$

$$\ddot{x} = \frac{1}{2}e^{4x}$$

Question 12 (e)

Criteria	Marks
• Provides correct solution	2
• Correctly uses double angle result, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2} \text{ since } \cos 2x = 1 - 2\sin^2 x \\
 &= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \\
 &= 2 \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \\
 &= 2 \times 1^2 \\
 &= 2
 \end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Correctly finds the value of n^2 , or equivalent merit	2
• Correctly uses $v^2 = n^2(a^2 - x^2)$, or equivalent merit	1

Sample answer:

$$v^2 = n^2(a^2 - x^2)$$

$$\therefore 4^2 = n^2 a^2 - 2^2 n^2 \quad \text{--- ①}$$

$$\text{and } 3^2 = n^2 a^2 - 5^2 n^2 \quad \text{--- ②}$$

$$\text{①} - \text{②} \quad \therefore 7 = -4n^2 + 25n^2$$

$$21n^2 = 7$$

$$n^2 = \frac{1}{3}$$

$$n = \frac{1}{\sqrt{3}}, \quad n > 0$$

$$\text{Period, } T = \frac{2\pi}{n}$$

$$= \frac{2\pi}{\frac{1}{\sqrt{3}}}$$

$$= 2\pi\sqrt{3}$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly applies binomial theorem to one binomial expression, or equivalent merit	1

Sample answer:

n is even

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{r}x^r + \cdots + \binom{n}{n}x^n$$

$$(1-x)^n = \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + (-1)^r \binom{n}{r}x^r + \cdots + \binom{n}{n}x^n$$

$$(1+x)^n + (1-x)^n = 2\binom{n}{0} + 2\binom{n}{2}x^2 + \cdots + 2\binom{n}{n}x^n$$

(odd terms have different sign and so cancel)

$$(1+x)^n + (1-x)^n = 2\left[\binom{n}{0} + \binom{n}{2}x^2 + \binom{n}{4}x^4 + \cdots + \binom{n}{n}x^n\right]$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Differentiating with respect to x .

$$n(1+x)^{n-1} - n(1-x)^{n-1} = 2\left[2\binom{n}{2}x + 4\binom{n}{4}x^3 + \cdots + n\binom{n}{n}x^{n-1}\right]$$

$$n((1+x)^{n-1} - (1-x)^{n-1}) = 2\left[2\binom{n}{2}x + 4\binom{n}{4}x^3 + \cdots + n\binom{n}{n}x^{n-1}\right]$$

Question 13 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Uses $x = 1$, or equivalent merit	1

Sample answer:

Let $x = 1$

$$n(2^{n-1} - 0) = 2 \left[2 \binom{n}{2} + 4 \binom{n}{4} + \cdots + n \binom{n}{n} \right]$$

$$n 2^{n-2} = 2 \binom{n}{2} + 4 \binom{n}{4} + \cdots + n \binom{n}{n}$$

$$n 2^{n-3} = \binom{n}{2} + 2 \binom{n}{4} + 3 \binom{n}{6} + \cdots + \frac{n}{2} \binom{n}{n}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $y = 0$, or equivalent merit	1

Sample answer:

$$\text{Set } 0 = y = Vt \sin \theta - \frac{1}{2}gt^2$$

$$t = 0 \quad \text{or} \quad V \sin \theta = \frac{1}{2}gt$$

$$t = \frac{2V \sin \theta}{g}$$

At this value

$$x = V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$$

$$= \frac{V^2}{g} \sin 2\theta$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\text{Range} = \frac{V^2}{g} \sin 2\theta$$

$$\text{If } v^2 < 100g, \text{ then range } \frac{100g \sin 2\theta}{g}$$

$$\text{ie } \text{range} < 100 \sin 2\theta$$

$$\text{since } \sin 2\theta \leq 1$$

$$\text{then } \text{range} < 100$$

Question 13 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Provides an inequality in $\sin 2\theta$, or equivalent merit	1

Sample answer:

$$\text{If } V^2 = 200g, \text{ then range} = \frac{200g \sin 2\theta}{g} \geq 100$$

$$\text{ie } 200 \sin 2\theta \geq 100$$

$$\sin 2\theta \geq \frac{1}{2}$$

$$\text{Now, } 0 \leq \theta \leq \frac{\pi}{2} \text{ or } 0 \leq 2\theta \leq \pi$$

$$\frac{\pi}{6} \leq 2\theta \leq \frac{5\pi}{6}$$

$$\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$$

Question 13 (c) (iv)

Criteria	Marks
• Provides correct solution	2
• Obtains correct expression for the maximum height, or equivalent merit	1

Sample answer:

Maximum height occurs when

$$0 = \frac{dy}{dt} = V \sin \theta - gt$$

$$t = \frac{V}{g} \sin \theta$$

At this time

$$y = \frac{V^2}{g} \sin^2 \theta - \frac{1}{2} g \frac{V^2}{g^2} \sin^2 \theta$$

$$= \frac{V^2 \sin^2 \theta}{g} \frac{1}{2}$$

$$= 100 \sin^2 \theta \quad \left(\frac{V^2}{g} = 200 \right)$$

For $0 \leq \theta \leq \frac{\pi}{2}$ $\sin^2 \theta$ is an increasing function,

so maximum occurs when $\theta = \frac{5\pi}{12}$

$\therefore$ greatest height is $100 \sin^2 \left(\frac{5\pi}{12} \right)$

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Correctly shows $P(n) \Rightarrow P(n+1)$, or equivalent merit	2
• Establishes result for $n = 1$, or equivalent merit	1

Sample answer:

Let $P(n)$ be the given proposition.

$P(1)$ is true since $8^3 + 6 = 518 = 7 \times 74$ which is divisible by 7.

Let k be an integer for which $P(k)$ is true.

That is $8^{2k+1} + 6^{2k-1} = 7m$, for some integer m .

Consider $8^{2(k+1)+1} + 6^{2(k+1)-1}$

$$= 8^2 \times 8^{2k+1} + 6^2 \times 6^{2k-1}$$

$$= 8^2 \times 8^{2k+1} + 8^2 \times 6^{2k-1} - 8^2 \times 6^{2k-1} + 6^2 \times 6^{2k-1}$$

$$= 8^2(8^{2k+1} + 6^{2k-1}) + (6^2 - 8^2)6^{2k-1}$$

$$= 64 \times 7m - 28 \times 6^{2k-1}$$

$$= 7(64m - 4 \times 6^{2k-1})$$

which is divisible by 7, since m and k are integers and $64m - 4 \times 6^{2k-1}$ is an integer.

$\therefore P(k+1)$ is also true

$\therefore P(n)$ is true for all $n \geq 1$ by induction.

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains equation of tangent at P , or equivalent merit	1

Sample answer:

The equation of the tangent at P is

$$y = px - p^2$$

substituting into $x^2 = -4ay$

$$-\frac{x^2}{4a} = px - p^2$$

$$\therefore x^2 + 4apx - 4ap^2 = 0$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct coordinates	2
• Correctly verifies one coordinate	1

Sample answer:

The sum of the roots of the equation is $-4ap$.

x -coordinate of M is average of the roots

$\therefore x$ -coordinate of M is $-2ap$.

Substituting into the tangent,

$$y = p \times -2ap - p^2$$

$$= -p^2(2a + 1)$$

$\therefore M$ is the point $(-2ap, -p^2(2a + 1))$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Eliminates the parameter, or equivalent merit	1

Sample answer:

$$\text{Let } x = -2ap, y = -p^2(2a + 1)$$

$$\begin{aligned}\therefore y &= -\left(\frac{x}{2a}\right)^2(2a + 1) \\ &= -x^2 \frac{(2a + 1)}{4a^2}\end{aligned}$$

$$\text{so } x^2 = -\frac{4a^2}{2a + 1}y$$

$$\text{Hence } \frac{a^2}{2a + 1} = 1$$

$$\text{ie } a^2 - 2a - 1 = 0$$

$$\therefore a = 1 \pm \sqrt{2}$$

$$\therefore a = 1 + \sqrt{2}, \text{ since } a > 0$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly applies the product rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{d}{dt}(F(t)e^{0.4t}) &= F'(t)e^{0.4t} + 0.4F(t)e^{0.4t} \\ &= e^{0.4t}(50e^{-0.5t} - 0.4F(t)) + 0.4F(t)e^{0.4t} \\ &= 50e^{-0.1t}\end{aligned}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly integrates the expression given in part (i), or equivalent merit	1

Sample answer:

Integrating,

$$\begin{aligned}F(t)e^{0.4t} &= \int 50e^{-0.1t} dt \\&= -500e^{-0.1t} + c\end{aligned}$$

Now $F(t) = 0$ when $t = 0 \therefore c = +500$

$$\therefore F(t)e^{0.4t} = 500(1 - e^{-0.1t})$$

$$F(t) = 500(e^{-0.4t} - e^{-0.5t})$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Makes substantial progress	1

Sample answer:

$$\begin{aligned}
 F'(t) &= 500(-0.4e^{-0.4t} + 0.5e^{-0.5t}) \\
 &= 0 \text{ for a maximum} \\
 \therefore 0.4e^{-0.4t} &= 0.5e^{-0.5t} \\
 \therefore \frac{5}{4} &= e^{0.1t} \\
 t &= 10 \ln \frac{5}{4} \\
 &(\approx 2.23 \text{ hours})
 \end{aligned}$$

Alternative

$$\begin{aligned}
 F'(t) &= 50e^{-0.5t} - 0.4F(t) \\
 &= 50e^{-0.5t} - 0.4[500(e^{-0.4t} - e^{-0.5t})] \\
 &= 250e^{-0.5t} - 200e^{-0.4t}
 \end{aligned}$$

For maximum $F'(t) = 0$

$$\begin{aligned}
 \therefore 250e^{-0.5t} &= 200e^{-0.4t} \\
 \frac{5}{4} &= \frac{e^{-0.4t}}{e^{-0.5t}} \\
 &= e^{0.1t} \\
 \therefore 0.1t &= \ln\left(\frac{5}{4}\right) \\
 t &= 10 \ln\left(\frac{5}{4}\right)
 \end{aligned}$$

2017 HSC Mathematics Extension 1

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	16.2	PE3
2	1	12.1	HE7
3	1	2.10	PE3
4	1	5.9	PE2
5	1	10.5	HE7
6	1	9.6	PE3
7	1	15.3	HE4
8	1	14.1	HE4
9	1	17.3	HE3
10	1	18.1	HE3

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	1	6.7	PE6
11 (b)	2	8.8, 15.5	HE4
11 (c)	3	1.4	PE3
11 (d)	2	15.3	HE4
11 (e)	3	11.5	HE6
11 (f)	1	13.6E	HE6
11 (g) (i)	1	18.2	HE3
11 (g) (ii)	1	18.2	HE3
11 (g) (iii)	1	18.2	HE3
12 (a)	2	2.8	PE3
12 (b) (i)	3	1.4, 1.2	PE6
12 (b) (ii)	1	1.4, 1.2	PE2
12 (c) (i)	3	11.4	H8, HE7
12 (c) (ii)	1	16.4	HE7
12 (d)	3	14.3	HE5
12 (e)	2	5.7, 8.2, 13.4E	HE7
13 (a)	3	14.4	HE3
13 (b) (i)	2	17.1, 17.3	HE7
13 (b) (ii)	1	17.1, 17.3	HE7
13 (b) (iii)	2	17.1, 17.3	HE7
13 (c) (i)	2	14.3	HE3
13 (c) (ii)	1	14.3	HE3
13 (c) (iii)	2	14.3	HE3

Question	Marks	Content	Syllabus outcomes
13 (c) (iv)	2	14.3	HE3
14 (a)	3	7.4	HE2
14 (b) (i)	2	9.6	PE3, PE4
14 (b) (ii)	2	9.6	PE3, PE4
14 (b) (iii)	2	9.6	PE3, PE4
14 (c) (i)	2	8.8, 11.2, 12.5	HE3
14 (c) (ii)	2	12.5	HE3
14 (c) (iii)	2	10.6, 12.5	HE3